

We argue that the biases are present both in the traditional specification of the gravity equation and in the Anderson-van Wincoop (2003) specification, which includes country-specific fixed effects.

To address the various estimation problems, we propose a simple Poisson pseudo-maximum likelihood method and assess its performance using Monte Carlo simulations. We find that in the presence of heteroskedasticity the standard methods can severely bias the estimated coefficients, casting doubt on previous empirical findings. Our method, instead, is robust to different patterns of heteroskedasticity and, in addition, provides a natural way to deal with zeroes in trade data.

We use our method to re-estimate the gravity equation and document significant differences from the results obtained using the log-linear method. For example, income elasticities in the traditional gravity equation are systematically smaller than those obtained with log-linearized OLS regressions. In addition, in both the traditional and Anderson-van Wincoop specifications of the gravity equation, OLS estimation exaggerates the role of geographical proximity and colonial ties. OLS estimates of the impact of preferential-trade agreements are very sensitive to the inclusion of fixed effects, whereas Poisson generates more stable results. RESET tests systematically favour the Poisson PML technique. The results suggest that heteroskedasticity (rather than truncation of the data) is responsible for the main differences.

Log-linearized equations estimated by OLS are of course used in many other areas of empirical economics and econometrics. Our Monte Carlo simulations and the regression outcomes indicate that in the presence of heteroskedasticity this practice can lead to significant biases. These results suggest that, at least when there is evidence of heteroskedasticity, the Poisson pseudo-maximum likelihood estimator should be used as a substitute for the standard log-linear model.

References

- Anderson, J. (1979). "A Theoretical Foundation for the Gravity Equation," *American Economic Review*, 69, 106-116.
- Anderson, J. and E. van Wincoop (2003). "Gravity with Gravitas: A Solution to the Border Puzzle," *American Economic Review*, 93, 170-192.
- Bergstrand, J. (1985). "The Gravity Equation in International Trade: Some Microeconomic Foundations and Empirical Evidence," *Review of Economics and Statistics*, 69, 474-481.
- Boisso, D. and M. Ferrantino (1997). "Economic and Cultural Distance in International Trade: Empirical Puzzles," *Journal of Economic Integration*, 12, 456-84.
- Cameron, A.C. and P.K. Trivedi (1998). *Regression Analysis of Count Data*, Cambridge, MA: Cambridge University Press.
- Davidson, R. and J.G. MacKinnon (1993). *Estimation and Inference in Econometrics*. Oxford: Oxford University Press.
- Davis, D. (1995). "Intra-industry Trade: A Heckscher-Ohlin-Ricardo Approach," *Journal of International Economics*, 39, 201-226.
- Deardoff, A. (1998). "Determinants of Bilateral Trade: Does Gravity Work in a Neoclassical World?" in Jeffrey Frankel (ed.), *The Regionalization of the World Economy*. Chicago, IL: University of Chicago Press.
- Delgado, M. (1992). "Semiparametric Generalized Least Squares Estimation in the Multivariate Nonlinear Regression Model," *Econometric Theory*, 8, 203-222.
- Delgado, M. and T.J. Kniesner (1997). "Semiparametric Versus Parametric Count-data Models: Econometric Considerations and Estimates of a Hedonic-Equilibrium Model of Workers Absentism," *Review of Economics and Statistics*, 79, 41-49.
- Eaton, J. and S. Kortum (2001). "Technology, Geography and Trade," NBER Working Paper No. 6253.
- Eaton, J. and A. Tamura (1994). "Bilateralism and Regionalism in Japanese and US Trade and Direct Foreign Investment Patterns," *Journal of the Japanese and International Economics*, 8, 478-510.

- Eichengreen, B. and D. Irwin (1995). "Trade Blocs, Currency Blocs, and the Reorientation of World Trade in the 1930's," *Journal of International Economics*, 38, 1-24.
- Eicker, F. (1963). "Asymptotic Normality and Consistency of the Least Squares Estimators for Families of Linear Regressions," *The Annals of Mathematical Statistics*, 34, 447-456.
- Feenstra, R.C., R.E. Lipsey and H.P. Bowen (1997), "World Trade Flows, 1970-1992, with Production and Tariff Data," NBER Working Paper No. 5910.
- Feenstra, R., J. Markusen and A. Rose (1999). "Using the Gravity Equation to Differentiate Among Alternative Theories of Trade," <http://www.econ.ucdavis.edu/~feenstra>.
- Frankel, J. (1997), *Regional Trading Blocs in the World Economic System*. Washington, DC: Institute for International Economics.
- Frankel, J. and A. Rose (2002). "An Estimate of the Effect of Common Currencies on Trade and Income," *Quarterly Journal of Economics*, 117, 409-466.
- Frankel, J. and S. Wei (1993). "Trade Blocs and Currency Blocs," NBER Working Paper No. 4335.
- Gourieroux, C., A. Monfort and A. Trognon (1984). "Pseudo Maximum Likelihood Methods: Applications to Poisson Models," *Econometrica*, 52, 701-720.
- Goldberger, A. (1968). "The Interpretation and Estimation of Cobb-Douglas Functions," *Econometrica*, 36, 464-472.
- Goldberger, A. (1991). *A Course in Econometrics*. Cambridge, MA: Harvard University Press.
- Gray, A. (2001). <http://argray.fateback.com/dist/formula.html>.
- Harrigan, J. (1993). "OECD Imports and Trade Barriers in 1983," *Journal of International Economics*, 35, 95-111.
- Haveman, J. and D. Hummels (2001). "Alternative Hypotheses and the Volume of Trade: The Gravity Equation and the Extent of Specialization," mimeo Purdue University.
- Helpman, E. and P. Krugman (1985). *Market Structure and Foreign Trade*. Cambridge, MA: MIT Press

- Koenker, R. and G.S. Bassett Jr. (1978). "Regression Quantiles," *Econometrica*, 46, 33-50.
- Manning, W.G. and J. Mullahy (2001). "Estimating Log Models: To Transform or Not to Transform?," *Journal of Health Economics* 20, 461-494.
- McCallum, J. (1995). "National Borders Matter: Canada-US Regional Trade Patterns," *American Economic Review*, 85, 615-623.
- McCullagh, P. and J.A. Nelder (1989). *Generalized Linear Models*, 2nd ed., London: Chapman and Hall.
- Papke, L.E. and J.M. Wooldridge (1996). "Econometric Methods for Fractional Response Variables with an Application to 401(k) Plan Participation Rates," *Journal of Applied Econometrics*, 11, 619-632.
- Park, R. (1966). "Estimation with Heteroskedastic Error Terms," *Econometrica*, 34, 888.
- Ramsey, J.B. (1969). "Tests for Specification Errors in Classical Linear Least Squares Regression Analysis," *Journal of the Royal Statistical Society B*, 31, 350-371.
- Robinson, P.M. (1987). "Asymptotically Efficient Estimation in the Presence of Heteroskedasticity of Unknown Form," *Econometrica*, 55, 875-891.
- Rose, A. (2000). "One Money One Market: Estimating the Effect of Common Currencies on Trade," *Economic Policy*, 15, 7-46.
- Tenreyro, S. and R. Barro (2002). "Economic Effects of Currency Unions," FRB Boston Series, Working Paper No. 02-4.
- Tinbergen, J. (1962). *The World Economy. Suggestions for an International Economic Policy*. New York, NY: Twentieth Century Fund.
- Wei, S. (1996), "Intra-National versus International Trade: How Stubborn Are Nation States in Globalization?," NBER Working Paper No. 5331.
- White, H. (1980). "A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity," *Econometrica*, 48, 817-838.
- Winkelmann, R. (2003). *Econometric Analysis of Count Data*, 4th ed. Berlin: Springer-Verlag.